

Palestine Ahliya University
Faculty of Graduate Studies
Machine Learning
Assignment 1

Instructor: Mutaz Abu Sara

Student: Andres Emaya

Tackling the Imbalanced Fraud Detection Dataset Using Various Techniques

Introduction

One of the most challenging issues facing machine learning experts in today's day and age is dealing with imbalanced datasets. The imbalance in the data occurs when one class overwhelms another class with the amount of either positives or negatives inside the class. This issue is persistent in cases where an event is unlikely to happen in comparison to a different event. In this report, we will plan to evaluate the Credit Card Fraud Detection by MLG and ULB. What makes this Dataset stand out is the amount of fraud that is detected. A very small percentage of the cases is considered fraud, while the rest are normal transactions. As a result, any naive model would give an accuracy of 99% or more on this particular dataset. This report presents the problems that imbalanced datasets impose on machine learning experts and using multiple strategies to handle such cases were implemented, trialed, and tested.

Strategies that were used include: SMOTE, handling outliers, data cleaning, feature extraction, scaling using the robust scaling method, and Ensemble learning to make sure that we select a strong model. Evaluation methods included ROC-AUC, F1 scores, recall, and precision. Accuracy would be irrelevant, since we are dealing with an imbalanced dataset. The models that have been used are LightGBM, XGBoost, Logistic Regression, and ensemble learning techniques.

The Data Set

The Credit Card Fraud data set consists of 284807 samples, and 31 features. Out of all the samples, only 492 samples were fraudulent. Without handling imbalances, the model would become bias towards the class that has the bigger number of samples, while simultaneously ignoring the class that has the smaller number of samples. In this case, we are more interested in the F1 score, recall, and precision, which gives us an insight on how our model is performing, unlike the accuracy, which gives a false 99.5% score.

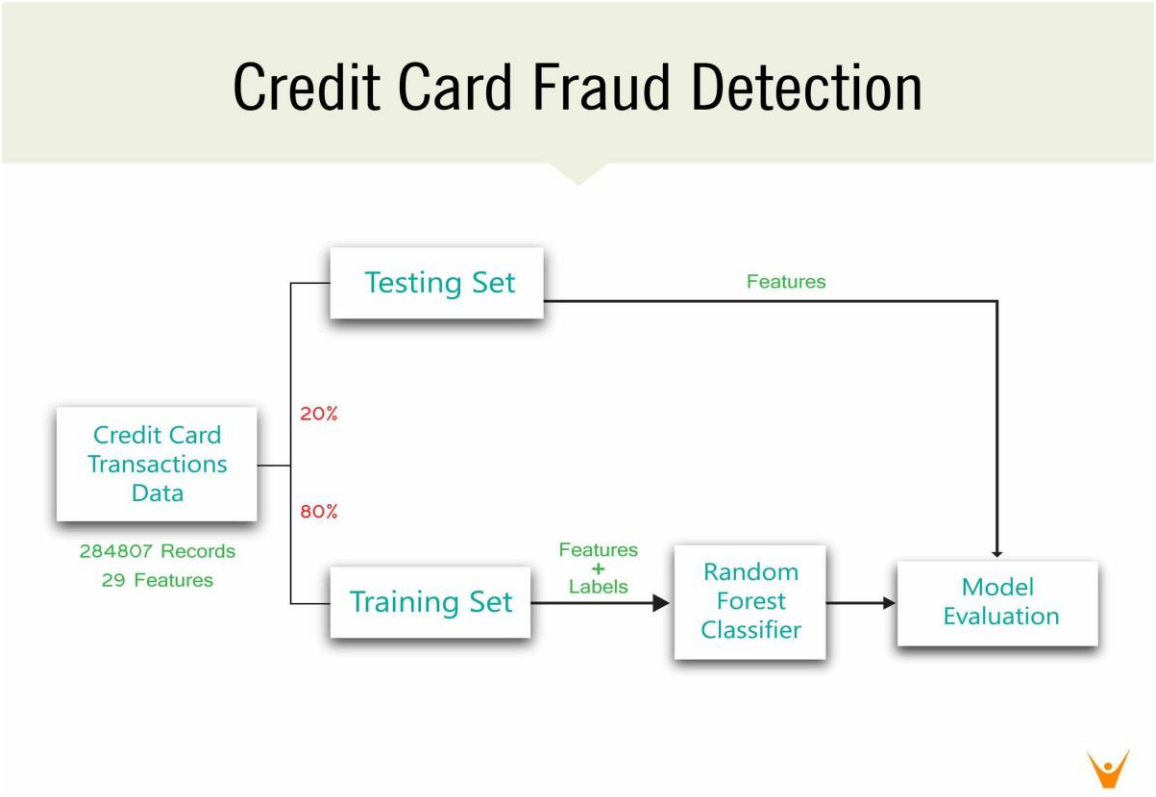


Figure 1: The Credit Card Fraud Detection Flow Chart

We are dealing with a binary classification problem, which contain numerical features.

0 → Refers to the legitimate Transaction

1 → Refers to the Fraudulent Transaction

Data-Preprocessing

SMOTE (Synthetic Minority Over-Sampling) was used to create new artificial data points that help in removing the bias for the class that has a lot of samples. It is also a technique that is used to reduce any over-fitting by the different models used in this report. Data cleaning was employed to remove any noise or redundant samples. First, we examined the dataset by using print(df['Class'].value_counts()), which showed a severely imbalanced dataset. After fixing the imbalanced dataset using various techniques, outlier clipping methods were utilized to remove any extreme values and make the generalization process smoother and more robust. Additionally, some features were removed which were deemed unnecessary using the LightGBM feature importance scores, and removing the least important ones.

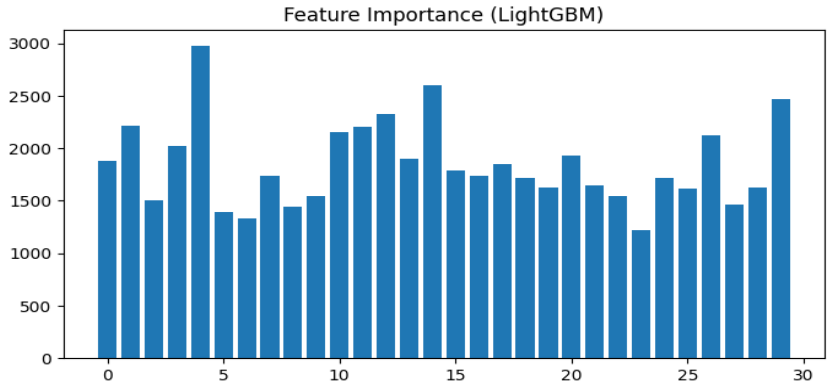


Figure 2: Feature Importance scores using LightGBM

As for the data splitting, the data was split using stratify, which preserves the class distribution, even after adding SMOTE and ensure a fair evaluation across all the possible splits. For the model we used a robust scaler, because it is not severely affected by outliers, or any skewed values. This makes it more resilient and robust for this particular dataset. After applying SMOTE and doing the necessary pre-processing, the distribution became as follows: [227451 56862]. It was expanded to accommodate for the severe imbalance in the dataset and to apply penalties to misclassifying the fraud cases. XGBOOST is used to improve the recall, which is the most important aspect of evaluation in this dataset. After applying all the abovementioned methods, Ensemble learning was employed to increase stability, robustness, and give a better generalization for new unseen data.

Results

Accuracy came up to 99.9%, which is expected, because any naïve model would get that score. We will not be focusing on accuracy in this report since it is irrelevant, rather we will focus more on other evaluation metrics.

The precision got a score of 94%, which is comparable to respected papers doing the same research on this dataset. The recall was 81%, which is 0.9% lower than the best score achieved on this particular dataset.

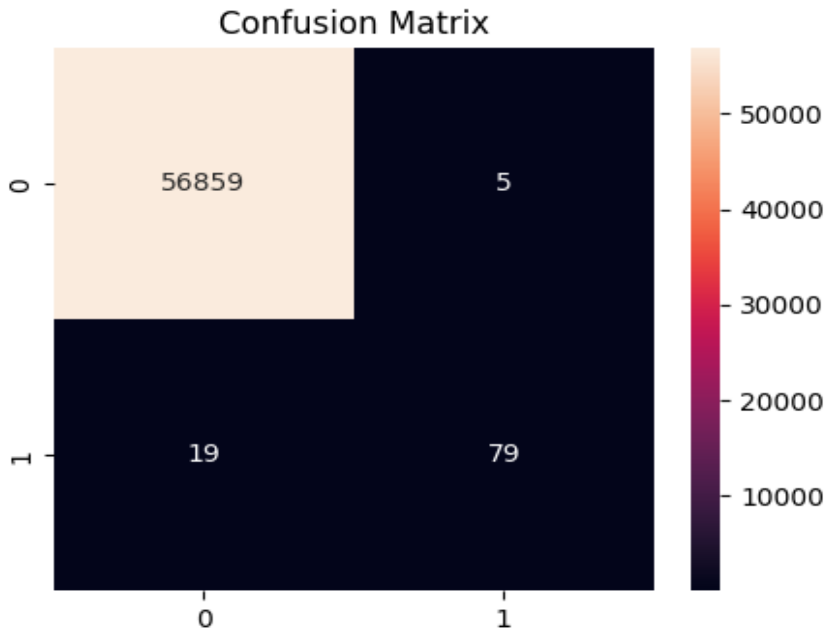


Figure 3: Confusion Matrix which explains the false positives and false negatives

The difference was that other papers used advanced deep learning models and techniques to boost the recall and precision scores more, and to make the dataset even more balanced. As we can see from my paper, we got respectable results only using machine learning techniques and SMOTE, which was particularly useful in making the dataset more balanced.

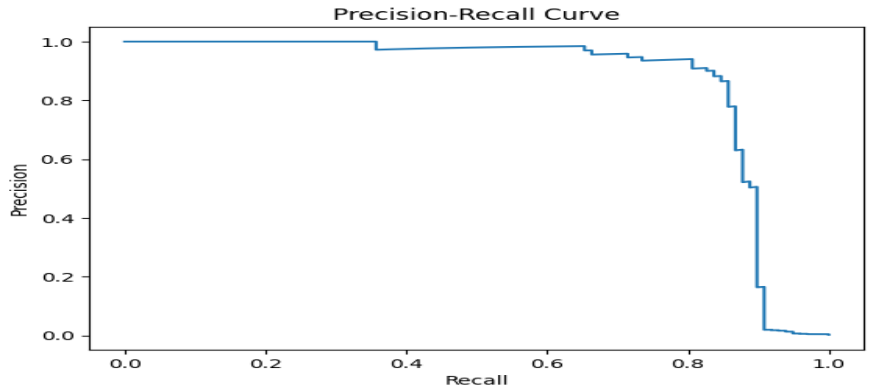


Figure 4: The Recall and Precision Curve

References

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). *SMOTE: Synthetic Minority Over-sampling Technique*. *Journal of Artificial Intelligence Research, 16*, 321–357. https://www.jair.org/index.php/jair/article/view/10302

He, H., & Garcia, E. A. (2009). *Learning from Imbalanced Data*. *IEEE Transactions on Knowledge and Data Engineering, 21*(9), 1263–1284. https://ieeexplore.ieee.org/document/4633969

Lemaître, G., Nogueira, F., & Aridas, C. K. (2017). *Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning*. *Journal of Machine Learning Research, 18*(17), 1–5. https://jmlr.org/papers/v18/16-365.html

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). *Scikit-learn: Machine Learning in Python*. *Journal of Machine Learning Research, 12*, 2825–2830. https://arxiv.org/abs/1201.0490

Liu, Z., Kang, J., Tong, H., & Chang, Y. (2021). *IMBENS: Ensemble Class-Imbalanced Learning in Python*. *arXiv preprint arXiv:2111.12776*. https://arxiv.org/abs/2111.12776